

Axle configuration & loading sheet

Shipment **RS-1042** | Tractor **T-101** | Semitrailer **V-101**
Fixture revision 1 - prepared September 11, 2026

Illustrative inputs only. This is not a scale ticket, manufacturer rating, legal clearance or evidence from a real carrier.

Empty coupled combination and configured limits

Group	Empty mass	Configured limit	Axles
Steer	5,000 kg	6,500 kg	1
Drive	5,000 kg	18,000 kg	2, equalized
Trailer	4,000 kg	14,000 kg	2, equalized

Configured gross limit: **40,000 kg**. Empty masses include the coupled tractor and trailer and must be added only once.

Geometry and cargo

Tractor steer-to-drive-group wheelbase	4.0 m
Fifth wheel from steer group	3.8 m
Kingpin-to-trailer-group span	10.0 m
Stored shipment cargo	32,000 lb
Placement A: cargo centre from kingpin	8.0 m
Placement B: same cargo moved toward kingpin	5.0 m

Review exercise

Calculate both placements without changing payload or group limits. Retain the failed placement, then record a separate assessment for the modeled repositioning. Physical repositioning is not demonstrated by this fixture. Dispatch approval remains a separate action.

Supported static model: one tractor and semitrailer, equalized axle groups, cargo within the kingpin-to-trailer-group span. No dynamic, tire, side-to-side or individual-axle measurement claims.